

Clinical trials for continuously monitored and updated AI systems

Wouter A. C. van Amsterdam, Michael Oberst, Jean Feng, Jenna Wiens, Shengpu Tang, Shalmali Joshi, Rajesh Ranganath, Mark Sendak, Uri Shalit, Julia E. Vogt, Brett Beaulieu-Jones, Muhammad Mamdani, David Kent, Patrick J. Heagerty, Thomas R. Fleming & Anna Goldenberg

 Check for updates

As artificial intelligence becomes embedded in clinical workflows, trials must accommodate ongoing monitoring and updates.

Artificial intelligence (AI) models are increasingly transitioning from experimental tools to active drivers of clinical care, prompting calls for rigorous clinical trials to evaluate their impact^{1,2}. Current trial guidance typically assumes static interventions with locked algorithms. However, an AI intervention is rarely a static, standalone model; it is a dynamic, socio-technical system integrated within health-IT infrastructure, user interfaces and clinical workflows³. A critical component of this system is ongoing performance monitoring and model refinement, which is essential for maintaining safety and effectiveness as the surrounding environment evolves^{4,5}. We argue that clinical trials must be designed to treat these monitoring and updating procedures as intrinsic to the intervention itself, ensuring that trial results accurately reflect real-world delivery.

At first glance, continuous model monitoring and updating may seem to be at odds with best practices for trial design. Standard practice dictates that the intervention should be fully prespecified⁶, with intermediate safety and efficacy data viewable only by an independent Data Monitoring Committee (DMC)⁷. These safeguards protect against arbitrary adaptation, which can lead to biased evaluation of trial endpoints, drops in compliance and uninterpretable treatment effect estimates⁸.

Consequently, existing guidelines on trials of AI systems such as SPIRIT-AI⁹ and CONSORT-AI¹⁰ focus on static AI algorithms, requiring the exact version of the AI algorithm to be prespecified. Most AI trials are also limited to evaluating static snapshots of this technology and, at best, allow study investigators to monitor operational indicators such as recruitment or data completeness^{11,12}, ignoring the fact that AI deployments are vastly more complex, dynamic systems in real-world practice.

The reality is that clinical trials have long allowed for dynamic interventions, if specific conditions are met¹³. For instance, implementation trials allow interventions such as community-based services to be tailored to local contexts, provided that the tailoring procedure is protocolized¹³. Similarly, trials of titrated therapies can allow dose adjustments based on patient response or adverse events, as long as a prespecified algorithm is followed¹⁴. These precedents suggest a viable baseline for extending clinical trial guidelines to encompass the complexity of continuously evolving AI interventions.

A roadmap for clinical trials of dynamic AI systems

The key to conducting clinical trials that are suitable for dynamic AI systems is to clarify the distinction between ‘monitoring as part of

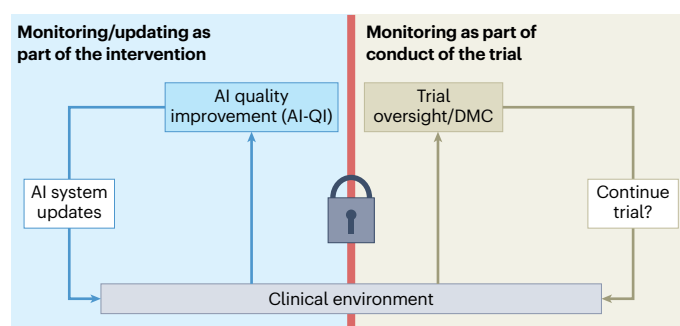


Fig. 1 | Conceptual framework separating monitoring and updating intrinsic to AI intervention delivery (left) from monitoring conducted as part of trial oversight (right). A governance boundary (lock) prevents trial data from influencing intervention behavior within the clinical environment.

the conduct of the trial’ and ‘monitoring and updating as part of the intervention’ (Fig. 1). Monitoring as part of the trial serves to protect the interests of trial participants while ensuring the integrity of the trial results, and is usually conducted by an independent DMC following a prespecified analysis plan.

Monitoring as part of the intervention refers to the continuous model monitoring and updating that are intrinsic to the AI intervention and thus persist beyond the trial throughout the lifespan of the tool being deployed. These protocols may rely on data generated during routine operation (such as alert frequency), as well as information external to the trial (such as routine vendor updates). In contrast, trial monitoring procedures typically rely on information that is available only in multi-arm studies, such as comparisons between treatment and control arms, and thus have no analogue in real-world deployment and cannot be considered as part of the intervention.

Critically, we argue that these two categories can and should coexist in a clinical trial so that the trial results are an unbiased reflection of real-world deployment settings, as long as appropriate guardrails are in place. Table 1 lists the potential risks that may be introduced when one considers allowing monitoring and updating as part of an AI intervention, along with potential mitigation strategies. An important part of these mitigation strategies is an independent clinical AI quality improvement (AI-QI) team⁴ composed of personnel (such as data scientists, IT and clinical personnel) who are responsible for monitoring and updating the AI system during the trial and once it is deployed after the trial.

We recognize that some of the proposed mitigation strategies may not be easy to implement in resource-limited settings. In such cases,

Table 1 | Recommendations when allowing both ‘monitoring as part of the conduct of a trial’ and ‘monitoring and/or updating as part of the intervention’

Potential risks of allowing both types of monitoring in a trial	Example mitigation strategies
Trial results that do not reflect real-world impact of the AI system because AI deployment during versus after the trial do not align	• Monitoring and updating procedures for the trial should follow prespecified post-deployment practices, such as outlined in Predetermined Change-Control Plans. • Personnel responsible for post-trial monitoring and updating (i.e., an AI-QI team) should also take responsibility for monitoring and updating during the trial. • The AI-QI team should be independent of trial investigators and AI developers to minimize conflicts of interest and differences in behavior during versus after the trial.
Trial signal leakage through operational monitoring, resulting in biased evaluation of trial endpoints and/or unintended impacts on user adoption	Compartmentalization of monitoring responsibilities; for example: • <i>Data Monitoring Committee (DMC)</i>: Exclusive access to comparative clinical outcomes. • <i>AI-QI team</i>: Access to system and workflow metrics (e.g., uptime, error rates, alert frequency, edit distance) that are unlikely to reveal comparative outcomes.
Lack of transparency due to the dynamic nature of the AI system	• Transparent documentation of any modifications of prompts, models or workflow integrations. • Prespecified update scope, version control and audit trails.

we suggest choosing a solution that is more feasible but respects the underlying principles. For example, it may not be possible for a hospital to assemble an independent AI-QI team that is distinct from the original AI developers and trial investigators. Nevertheless, the underlying principle is to minimize the risk of bias and conflicts of interest by avoiding situations in which the trial investigators are the same individuals responsible for monitoring and updating of the AI system.

We now describe two case studies to illustrate how the two categories can coexist without compromising the scientific integrity of a clinical trial.

Use case 1: detection of pulmonary embolism

Consider an AI system designed to detect pulmonary embolism (PE) on computed tomography (CT) scans. The system runs in the background, automatically analyzing every CT scan acquired in the hospital and generating alerts for the interpreting radiologist when indicators of PE are identified. To evaluate the system’s clinical impact, imagine a hypothetical multi-center cluster-randomized trial in which hospitals are randomized to either implement the AI system or continue with standard care; the primary endpoint is all-cause mortality and the secondary endpoint is serious adverse events (SAEs) related to anti-thrombotic treatments.

Monitoring as part of the trial. The DMC should have exclusive access to interim data on differences in deaths and SAEs between trial arms, because these comparisons exist only within the trial and, if inadvertently disclosed, could bias the conduct of trial investigators or participants. As part of its responsibility to protect patient safety and preserve the validity of the randomized comparison, the DMC may recommend pausing trial enrollment if reliable evidence emerges that the AI system has an unfavorable benefit-to-risk profile.

Monitoring as part of the intervention. In parallel, an independent AI-QI team may follow protocolized procedures to monitor the safe functioning of the system using measures such as server uptime and error logs, image acquisition quality, model outputs and human–AI interaction signals, for example radiologist agreement with alerts and user feedback. These signals should be tied to clear, prespecified actions, such as automated failover and IT notification for downtime as well as clearly defined retraining or recalibration processes to be performed if radiologist confirmation rates decline.

Importantly, the AI-QI team should not have access to interim data contrasting the two arms of the trial, because such data would not be available during real-world deployment. Furthermore, the AI-QI team should avoid monitoring the primary and secondary endpoints of the trial, even just within the exposure arm, as that may unintentionally influence how trial data is generated or collected and thereby threaten the scientific integrity of the trial. After the trial concludes, however, the team may use these endpoints as part of routine monitoring.

Use case 2: automated generation of discharge summaries

Commercial AI tools are increasingly being used to summarize prior clinical notes and draft documentation for patient care tasks. Here, we imagine a non-inferiority randomized trial evaluating an AI tool that drafts discharge summaries compared with the standard workflow of manual documentation. The primary endpoints are patient outcomes such as readmission and documentation quality, assessed for accuracy and completeness by independent clinical reviewers; secondary endpoints include hospital process measures such as time savings and reductions in self-reported burnout. This trial is effectively open label because clinicians and reviewers are likely to know whether an AI draft was used, and some endpoints are inherently subjective, increasing the risk that disclosure of intermediate results could introduce bias.

Monitoring as part of the trial. The DMC should retain sole responsibility for monitoring the effect of the AI intervention compared with standard care, because between-arm contrasts cannot be incorporated into ongoing monitoring as part of the intervention itself. In addition, even within the intervention arm, monitoring of documentation quality may need to remain trial specific if the assessment process depends on resource-intensive manual review that would not be feasible during routine clinical delivery.

Monitoring as part of the intervention. An AI-QI team independent of the trial investigators should generally be permitted to follow protocolized procedures for monitoring technical functionality, resolving implementation problems, updating user training in response to common questions or errors, and incorporating vendor-released model updates, including newer foundation models. These activities pose limited risk of bias when prespecified, but greater care is needed around monitoring serious summarization errors in AI drafts: although such information is less likely to affect objective endpoints such as readmission, broader disclosure could influence more subjective outcomes such as documentation quality and self-reported burnout. For that reason, intermediate error reports should be tightly compartmentalized to the AI-QI team, and data on primary clinical endpoints should remain exclusive to the DMC. Using external clinical reviewers may also reduce the risk of biased assessment of documentation quality.

Discussion

Clinical AI interventions are dynamic systems that require continuous monitoring and updating, and trials should evaluate them as such. Although some regulatory frameworks are already moving in this direction, such as the FDA's Predetermined Change-Control Plan guidance for AI/ML-based Software as a Medical Device, trial-specific operationalization remains absent from existing standards such as SPIRIT-AI and CONSORT-AI. Here we take an initial step toward addressing this gap by proposing a framework that distinguishes monitoring conducted for trial oversight from monitoring and updating intrinsic to AI intervention delivery.

Clarifying this boundary allows trials to preserve the rigor of randomized evaluation while accommodating the operational practices required to keep AI systems safe and effective in practice. Moving forward will require collaboration among trialists, regulators, methodologists, health systems and technology developers to develop governance templates and trial designs that institutions with varying resources can realistically adopt. Allowing continuous monitoring and updating of AI systems during clinical trials is essential for generating rigorous evidence about interventions whose real-world use involves ongoing monitoring and refinement.

Wouter A. C. van Amsterdam ^{1,18}, **Michael Oberst** ^{2,18}, **Jean Feng** ^{3,18}, **Jenna Wiens** ⁴, **Shengpu Tang** ⁵, **Shalmali Joshi**⁶, **Rajesh Ranganath** ⁷, **Mark Sendak** ⁸, **Uri Shalit**⁹, **Julia E. Vogt**¹⁰, **Brett Beaulieu-Jones** ¹¹, **Muhammad Mamdani**^{12,13,14}, **David Kent**¹⁵, **Patrick J. Heagerty**¹⁶, **Thomas R. Fleming** ¹⁶ & **Anna Goldenberg**^{13,14,17} 

¹Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht, the Netherlands. ²Johns Hopkins University, Baltimore, MD, USA. ³University of California, San Francisco, CA, USA. ⁴University of Michigan, Ann Arbor, MI, USA. ⁵Emory University, Atlanta, GA, USA. ⁶Columbia University, New York, NY, USA. ⁷New York University, New York, NY, USA. ⁸Vega Health, Durham, NC, USA. ⁹Tel Aviv University, Tel Aviv, Israel. ¹⁰ETH Zurich, Zurich,

Switzerland. ¹¹University of Chicago, Chicago, IL, USA. ¹²Ontario Health, Toronto, Ontario, Canada. ¹³University of Toronto, Toronto, Ontario, Canada. ¹⁴Vector Institute, Toronto, Ontario, Canada. ¹⁵Tufts University, Boston, MA, USA. ¹⁶University of Washington, Seattle, WA, USA. ¹⁷Hospital for Sick Children, Toronto, Ontario, Canada. ¹⁸These authors contributed equally: Wouter A. C. van Amsterdam, Michael Oberst, Jean Feng.

✉ e-mail: anna.goldenberg@utoronto.ca

Published online: 28 April 2026

References

1. Adams, R. et al. *Nat. Med.* **28**, 1455–1460 (2022).
2. Tai-Seale, M. et al. *JAMA Netw. Open* **7**, e246565 (2024).
3. Joshi, S. et al. *J. Am. Med. Inform. Assoc.* **32**, 589–594 (2025).
4. Feng, J. et al. *npj Digit. Med.* **5**, 1–9 (2022).
5. Finlayson, S. G. et al. *N. Engl. J. Med.* **385**, 283–286 (2021).
6. Chan, A.-W. et al. *BMJ* **346**, e7586 (2013).
7. Ellenberg, S. S., Fleming, T. R. & DeMets, D. L. *Data Monitoring Committees in Clinical Trials: A Practical Perspective* (John Wiley & Sons, 2019).
8. Fleming, T. R. et al. *Clin. Trials* **5**, 157–167 (2008).
9. Rivera, S. C. et al. *BMJ* **370**, m3210 (2020).
10. Liu, X. et al. *Nat Med* **26**, 1364–1374 (2020).
11. Wijnberge, M. et al. *JAMA* **323**, 1052–1060 (2020).
12. Angus, D. C. *JAMA* **323**, 1043–1045 (2020).
13. Kilbourne, A., Chinman, M., Rogal, S. & Almirall, D. *Annu. Rev. Public Health* **45**, 69–88 (2024).
14. ACCORD Study Group. et al. *N. Engl. J. Med.* **362**, 1575–1585 (2010).

Acknowledgements

Support for this work was provided by the Varma Family Chair in Biomedical Informatics and Artificial Intelligence and CIFAR AI Chair to A.G.; the US National Institutes of Health, R37-AI29168 to T.R.F.; the PatientCentered Outcomes Research Institute (PCORI) Award (ME-2022C125619) to J.F.; and a grant from the Agency for Healthcare Research and Quality (R01HS027431) to J.W. The views presented in this work are solely the responsibility of the author(s) and do not necessarily represent the views of the PCORI or its board of governors or methodology committee.

Competing interests

M.O. holds equity in Abridge AI and is employed in a part-time role as a visiting scientist at Abridge AI. U.S. is a visiting research scientist at Google. M.M. is an advisor to Signal 1 and has given talks sponsored by Takeda, Bayer and Apobiologix.